EQUIPMENT NAME:	**FIM III Booster Station LV Motor Control Centre**
EQUIPMENT NUMBER:	**F23-SW-602**

Procurement Package	"EL006" - 415V Motor Control Centre			
Equipment No.	**Equipment Name**	**Construction**	**Busbar Rating**	**Fault Withstand Current Rating**
F23-SW-602	FIM III Booster Station LV Motor Control Centre	Double Sided	4000A	65kA for 1 second

Reference Documents		
Equipment Lists		
Document Number	**Rev**	**Description**
2000-F23-LST-EL-40001	0	FIM III Booster Station Electrical Load List
2000-F23-LST-EL-40002	0	FIM III Booster Station Electrical Equipment List
2000-F23-LST-EL-40003	0	FIM III Booster Station Cable Schedule
2000-F23-LST-EL-40006	0	FIM III Booster Station LV Components List
Block Diagrams		
2000-F23-BLK-IC-40001	A	F23-SR-601 HV/LV Switchroom Communications Block Diagram
Single Line and Three-line Diagrams		
2000-F23-SLD-EL-40001	0	F23-SW-601 HV Switchboard Single Line Diagram
2000-F23-SLD-EL-40002	0	F23-SW-601 HV Switchboard Key Interlocking Single Line Diagram
2000-F23-SLD-EL-40011	1	F23-SW-602 LV Motor Control Centre Single Line Diagram Sheet 1 of 2
2000-F23-SLD-EL-40012	2	F23-SW-602 LV Motor Control Centre Single Line Diagram Sheet 2 of 2
2000-F23-TLD-EL-40001	0	F23-BC-601 Switchroom 48VDC Battery Charger Three-Line Diagram
2000-F23-TLD-EL-40011	1	F23-DB-601 Switchroom and Area L&SP DB Three-Line Diagram
2000-F23-TLD-EL-40012	1	F23-DB-602 Switchroom HVAC DB Three-Line Diagram
2000-F23-TLD-EL-40013	0	F23-DB-603 Switchroom 240VAC UPS DB Three-Line Diagram
Schematic, Termination and Connection Diagrams		
2000-F23-SCM-EL-40011	0	F23-SW-602 LV Motor Control Centre Incomer Schematic Diagram Sheet 1
2000-F23-SCM-EL-40012	0	F23-SW-602 LV Motor Control Centre Incomer Schematic Diagram Sheet 2
2000-F23-SCM-EL-40013	0	F23-SW-602 LV Motor Control Centre Arc Flash Connection Diagram
2000-F23-SCM-EL-40011	0	F23-SW-602 LV Motor Control Centre Incomer Schematic Diagram Sheet 1
2000-F23-SCM-EL-40012	0	F23-SW-602 LV Motor Control Centre Incomer Schematic Diagram Sheet 2
2000-F23-SCM-EL-40013	0	F23-SW-602 LV Motor Control Centre Arc Flash Connection Diagram
2000-F23-SCM-EL-40014	0	F23-SW-602 24VDC Distribution Schematic Diagram
2000-F23-SCM-EL-40015	0	F23-DB-601 Switchroom and Area L&SP DB Control Schematic Diagram
2000-F23-SCM-EL-40016	0	F23-DB-602 Switchroom HVAC 415V DB Control Schematic Diagram
2000-F23-SCM-EL-40017	0	F23UPS601-ISL 240VAC UPS Maintenance Bypass Switch Control Schematic Diagram
2000-F23-SCM-EL-40018	0	F23-PP-651 FIM III Booster Pump 1 VSD 415V Auxiliary Supply Control Schematic Diagram
2000-F23-SCM-EL-40019	0	F23-PP-652 FIM III Booster Pump 2 VSD 415V Auxiliary Supply Control Schematic Diagram
2000-F23-SCM-EL-40020	0	F23-PW-651 FIM III Booster Pump 1 Seal Water Pump 1 Control Schematic Diagram
2000-F23-SCM-EL-40021	0	F23-PW-652 FIM III Booster Pump 1 Seal Water Pump 2 Control Schematic Diagram
2000-F23-SCM-EL-40022	0	F23-PW-653 FIM III Booster Pump 2 Seal Water Pump 1 Control Schematic Diagram
2000-F23-SCM-EL-40023	0	F23-PW-654 FIM III Booster Pump 2 Seal Water Pump 2 Control Schematic Diagram
2000-F23-SCM-EL-40024	0	F75-FL-600 FIM III Booster Pumps Seal Water Self-Cleaning Filter Control Schematic Diagram
2000-F23-SCM-EL-40025	1	F23-HY-420 FIM III Hydraulic Power Pack Control Schematic Diagram
2000-F23-SCM-EL-40026	1	F23-FN-653 FIM III Booster Pump 1 Motor Fan Control Schematic Diagram
2000-F23-SCM-EL-40027	1	F23-FN-654 FIM III Booster Pump 2 Motor Fan Control Schematic Diagram
2000-F23-SCM-EL-40028	1	F23-FN-651 FIM III Booster Pump 1 Gearbox Fan Control Schematic Diagram
2000-F23-SCM-EL-40029	1	F23-FN-652 FIM III Booster Pump 2 Gearbox Fan Control Schematic Diagram
2000-F23-SCM-EL-40030	0	F23-SE-431 Safety Shower Control Schematic Diagram
2000-F23-SCM-EL-40031	1	F23-HF-601 LV MCC Active Harmonic Filter Feeder Control Schematic Diagram
2000-F23-TER-EL-40011	0	F23-SW-602 LV Motor Control Centre Incomer Termination Diagram Sheet 1
2000-F23-TER-EL-40012	0	F23-SW-602 LV Motor Control Centre Incomer Termination Diagram Sheet 2
2000-F75-SCM-EL-40001	0	F75-PW-606 Decant Dam 4 Return Pump 1 Control Schematic Diagram
2000-F75-SCM-EL-40002	0	F75-PW-607 Decant Dam 4 Return Pump 2 Control Schematic Diagram
2000-F75-SCM-EL-40003	0	F75-PW-608 Decant Dam 4 Return Pump 3 Control Schematic Diagram
2000-F75-SCM-EL-40004	0	F75-PV-414 Decant Dam 4 Vacuum Pump Control Schematic Diagram
2000-F75-SCM-EL-40005	0	F75-PW-665 Washdown Pump Control Schematic Diagram

Reference Specifications
2000-F00-STS-EL-10014_2 - Preferred Equipment Suppliers for Electrical, Instrumentation and Controls Specification
2000-F00-STS-EL-10016_0 - Low Voltage Switchboards & Switchgear Electrical Specification
2000-F00-STS-GE-10002_0 - Site Specific Information
2000-F00-STS-GE-10003_2 - Plant Numbering Specification
2000-F00-STS-ST-10001_2 - Steelwork Fabrication
2000-F00-STS-ST-10002_2 - Steelwork Erection Specification
SPEC-00-G-05_12 – Technical Specification for Protective Coatings

Scope of Work

Work Included

- Design, supply and fabrication of one (1) indoor motor control centre including, but not limited to:
  - Withdrawable Form 4a Incomer ACB;
  - Demountable Form 4a Outgoing Modules;
  - All internal MCC Components including circuit breakers, E/L circuit breakers, contactors, electronic overloads (EOLs), thermal overloads (TOLs), soft starters, relays, transformers, UPS 240VAC to 24VDC Power supplies and Distribution, terminals, shrouds etc;
  - CAT6 wiring to intelligent electronic devices (IEDs) (EOLs, soft starters, relays, etc).
  - Arc fault detection at incomer and all outgoing modules including cable ways. The arc fault detection for the incomer shall be current supervised to trip the incomer and upstream feeder using both current and optical sensors.
  - All internal wiring and cabling including power, control, copper communications and fibre communications;
  - Labelling, including stainless steel equipment nameplate;
  - Lifting lugs, as applicable;
  - Any other items which are normally supplied with this equipment;
- Surface Protection;
- All ancillary items and special tools normally supplied with, or required to adjust and maintain all components associated with the equipment;
- All fasteners, gaskets, packers, shims and any other items necessary for site assembly;
- Factory Acceptance Testing at manufacturer's premises;
- Supply of all necessary documentation, drawings and test documentation;
- Supplier shall provide a schedule of Take In and Take Out pricing for each size of MCC module and for a whole tier;

Special Notes:

- The LV MCC package will be awarded together with the switchroom package "EL006" to the successful tender.

Work Excluded

- Cabling external to switchboard and switchroom;
- Spare parts;
- Installation and commissioning at site

Design Notes

Engineer's document package includes preliminary drawings and lists. For Construction documents will be produced following award of contract, if successful. 24V DC distribution schematic will be updated to suit MCC GA following receipt of successful vendor's approved MCC GA drawings.

Supplier shall nominate suitable circuit breaker sizes to suit the type 2 high efficiency coordination tables for the offered switchgear. Multi-motor drives shall ensure that electronic overloads and their associated contactors are type 2 high efficiency coordinated to suit the individual motors and shall be suitable for the full load current, noting that vibrating motors (such as those used in screens) typically have a larger full load current than a standard motor.

DOL motor starter modules will utilise electronic overloads with Modbus TCP communications functionality. Protection functions required from the overload are stated further in this datasheet. Soft Starters, where utilised, will have ethernet communications. IO Expansion modules may be utilised where required (e.g. multi-motor DOL).
MCC incomer arc flash detection scheme protection CTs are to be mounted on the upstream transformer by Vendor.

24V DC distribution schematic describes indicative 24V DC distribution throughout the MCC. It is envisaged that an MCB will be installed at the top of each cable zone. 24V DC control supply will be looped between DC distribution compartment and MCBs at top of tier. Within each tier, 24V DC control will loop from the MCB at top of tier to each individual starter module. Supplier shall ensure that their offer includes all necessary equipment for the 24V DC distribution.

Live line indication and earth leakage shall be provided for all incomers and MCC modules. Other forms of imbalance or earth leakage protection required for all hardwired sub-circuits.

Arc fault detection shall be provided at incomer and all outgoing modules including cable ways via optical sensors. The arc fault detection for the incomer shall be current supervised to trip the incomer and upstream feeder using both current and optical sensors. Arc fault detection current transformers to be installed in the transformer room by the Switchroom Vendor.

Description	Unit	Duty	Vendor	Notes	Rev
General					
Equipment Number		F23-SW-602		1	0
Equipment Name		FIM III Booster Station LV Motor Control Centre		1	0
Make			POWINS	2	0
Model			Ecoblock Plus	2	0
MCC Type		Air Insulated, Metal Clad, Form 4a, Free Standing, Double Sided		1	0
ACB Functional Unit Connections		Withdrawable		1	0
Non-ACB Functional Unit Connections (AS61439)		Demountable		1	0
Country of Manufacture			Malaysia /Australia	2	0
Compliance with Australian Standards	Yes/No	Yes AS60947, AS61439		1, 6	0
Drive and Load Descriptions		As per lists		1	0
Environment Conditions					

Description	Unit	Duty	Vendor	Notes	Rev
Plant Location		Western Australia		1	0
Service Location	Indoor/Outdoor	Indoor		1	0
Site Data		Refer to 2000-F00-STS-GE-10002 – Site Specific Information		1	0
Supply					
Supply		3P + N + E		1	0
Input Voltage	V	415V +/- 10%		1	0
Frequency	Hz	50 (±2%)		1	0
Neutral Earthing		Solid – M.E.N. on MCC (TN-S)		1	0
Switchboard Ratings					
Type Tested Assembly to AS61439	No/Partially/Fully	Fully Type Tested Assembly		1, 6	0
Switchboard Rating Plate		Yes		1	0
Main Bus Bar Continuous Current Rating	A	4000A	Confirmed	3	0
Neutral Bus Bar Continuous Current Rating	A	100% of phase rating	Confirmed	3	0
Main & Neutral Bus Bars Fault Withstand Level	kA for sec	Minimum 65kA for 1 Second	Confirmed	3	0
Dropper Bus Bars Fault Withstand Level	kA for sec	Minimum 65kA for 1 Second	Confirmed	3	0
Earth Bus Bars Fault Withstand Level	kA for sec			2	0
Rated Insulation Voltage	V	1000V	Confirmed	3	0
Interruption Capacity (All circuits connected directly and indirectly to main bus including ACBs, MCCB and 240V control circuit protection). ** Cascading is not permitted, fault current limiting fuses to be used where required.	kA	65kA Icu Vendor to Advise kA Ics (ACB incomer shall be rated to 65kA Icu/Ics)	Confirmed	3	0
Incomer Current Rating (de-rated according to manufacturer's de-rating tables and enclosure)	A	4000A ACB derated to 3600A as per type tests	Confirmed	3	0
Internal Arc Fault Containment (IAC) Rating to IEC TR 61641 / AS 61439.1 Annex ZD - Incomer Line Side Terminals - Main & Dropper Bus Bar System Outgoing Modules	kA for sec	Supplier to confirm ratings below: 1. 63kA/1s minimum 2. IAC rated bus not required if insulated 63kA/1s minimum Outgoing modules to AS61439.1 Annex ZD	Confirmed	3	0
Cascading used to achieve minimum breaking capacity	Yes / No	No		1	0
Switchboard Construction					
General Construction		Free Standing Metal Clad Air Insulated	Confirmed	3	0
Plinth Construction			HDG Steel	2	0
Degree of Ingress Protection	IP	Min IP42	Confirmed	3	0
Type of Cooling		Air cooled (arc contained non ventilated)	Confirmed	1	0
Exterior Paint Colour		AS2700 Tropical Blue T11		1	0
Interior Paint Colour		AS2700 / White		1	0
Paint Specification		Vendor Standard	Confirmed	2	0
Single Sided / Double Sided	Single / Double	Double Sided		1	0
Spare Space Allowance for Future Outgoing Modules		One spare double-sided functional tier and cable zone tier with no module installed on the end the MCC	Confirmed	3	0
Capacity for Future Extension	Yes / No	Yes, at one end		1	0
Switchboard Incomer Form		Form 4a		1	0
Switchboard Outgoing Motor Starters and Feeders Form		Form 4a		1	0
Cable Entry	Top / Bottom	Bottom Entry		1	0
Drop Boxes	Yes / No	Yes, with side panels to allow single person Sealing of cables from below (part of switchroom contract)		1	0
Seal Plates		5mm aluminium, earthed with neoprene gasket		1	0
Lifting Lugs			Lifting through HDG Steel Plinth	2	0
Jacking Beam Lugs		No – Base of switchboard must be vermin proof and IP rated.		1	0
Busbars					
Phase Busbar Size	mm^2		3 x 10 x 150	2	0
Neutral Busbar Size	mm^2		3 x 10 x 150	2	0

Description	Unit	Duty	Vendor	Notes	Rev
Earth Busbar Size	mm^2		1 x 6 x 6	2	0
Phase Dropper Size	mm^2		6 x 60 minimum	2	0
Neutral Dropper Size	mm^2		6 x 60 minimum	2	0
Busbar Identification		Colour Coded (Red, White, Blue Phases, Black Neutral & Green/Yellow Earth)		1	0
Busbar Material		HDHC Copper		1	0
Fully Insulated Bus Bars and Droppers	Yes / No	Droppers fully segregated	Confirmed	3	0
Main Busbar Location / Accessibility		Top – 2front accessible IP6X metal clad segregation	Confirmed	3	0
Phase and Wire Colours					
Phase Colours		Red, White, Blue		1	0
Neutral Colour		Black		1	0
Earth Colour		Green / Yellow		1	0
24V DC +24V DC (+ve)		Orange		1	0
24V DC 0V (-ve)		Purple		1	0
Analog 4-20mA		Pink		1	0
Ethernet Cables		Grey Sheath		1	0
Fibre Optic Patch Leads		Yellow – Single Mode		1	0
Live Line Indication					
Manufacturer		Nilsen	Remlive	3	0
Model		NWA2019	RL24I	3	0
Incomer Cubicle		Yes – Indicator mounted in control compartment		1	0
Incomer Cubicle Live Line Sensing		Pick up from line side of ACB, as per schematic		1	0
Outgoing Modules		Yes – Indicator mounted on module door. Pick up from downstream side of MCCB, as per schematics		1	0
Locking					
Incomer Padlockable in Withdrawn Position	Yes / No	Yes, 6.3mm Shackle		1	0
Outgoing MCCB comes with mechanically interlocked cubicle door mounted rotary handles so that: • The door cannot be opened in the on position. • The circuit breaker rotary handle can be locked in the off position and the door cannot be opened while padlocked off. • The circuit breaker extension handle shall be lockable inside the module allowing the MCCB to be locked out while the door is open. • The circuit breaker rotary handle cannot be padlocked in the on position.	Yes / No	Yes, 6.3mm Shackle		1	0
Incomer ACB					
Incomer Type		3P Withdrawable ACB, padlockable in withdrawn position		1	0
Fixed / Demountable / Withdrawable		Withdrawable (motorised)		1	0
Incomer ACB Make		NHP Terasaki		1	0
Incomer ACB Model		TemPower2 AR series	Confirmed	3	0
Shunt and Trip Coils		Yes 24 VDC		1	0
Motor Operated Spring Charge		Yes 240 VAC		1	0
External Neutral CT (installed in MCC)		Yes	Confirmed	3	0
Motorised Racking		Yes, for remote racking utilizing TemRack iR	Confirmed	3	0
Electronic Trip Unit		Yes – AGR-31C Adjustable phase overload time delay curve, definite time and instantaneous protection. Independent Neutral protection. Ground fault Protection. Integral Metering.	Confirmed	3	0
ACB Communications		Via Modbus TCP Communication module. Capable to remote Open/ Close via Modbus TCP.		1	0
Incomer ACB Status Contacts		Circuit breaker open (2 off), Circuit breaker closed (2 off), Circuit breaker tripped (1 off), Earth Fault Trip (1 off)		1	0
Auxiliary Supply		Yes 24 VDC. Trip unit shall be self-powered and auxiliary 24 VDC powered.		1	0
Remote Control Panel		Vendor to make allowance for IO to and from NHP's TemRack IR Local Control Station.		1	0

Description	Unit	Duty	Vendor	Notes	Rev
Earth Fault Relay					
Manufacturer		NHP	Confirmed	3, 7	0
Model		RD1EP21H + compatible NHP TD toroid	Confirmed	3, 7	0
Surge Diverters					
Surge Diverters required on Incomer	Yes/No	Yes		1	0
Manufacturer	-	Novaris	Confirmed	3	0
Model	-	SD3-200-275	Confirmed	3	0
Minimum and Maximum Fuse Size	-	63A	Confirmed	3	0
Provides Surge Diverter Status Contact to DCS	Yes / No	Yes		1	0
Current Transformers					
Current Transformers – ACB line side					
Manufacturer		Stemar/Ampcontrol	Confirmed	3	0
Type	Block / Toroid	Toroid		1	0
Duty		Protection Class CT – Arc Detection System (supplied loose for installation in transformer room) Metering Class CT – Power Quality Meter		1	**0**
Ratio		Refer to Busbar Ratings in Table above		1	0
Class		5P20 2.5VA (Protection) Cl 1.0 5VA (Metering)		1	0
Active Harmonic Filter Current Transformers					
Manufacturer			Stemar	2	0
Type	Block / Toroid	Toroid		1	0
Duty		Active harmonic Filter		1	0
Ratio		As per manufacturer requirements.	TBA	3	0
Class		As per manufacturer requirements.	TBA	3	0
MEN Link Current Transformer					
Manufacturer		NHP	Confirmed	3	**0**
Type	Block / Toroid	Toroid		1	0
Duty		Earth Leakage Protection via external Earth Leakage Relay		1	0
Ratio		As per manufacturer requirements.	TBA	3	0
Class		As per manufacturer requirements.	TBA	3	0
Power Quality Meter					
Manufacturer		SEL	Confirmed	3	0
Model		735#824G	Confirmed	3	0
Power Supply		240 VAC	Confirmed	3	0
Communications		Via Modbus TCP Communication Module		1	0
Arc Fault Detection System					
Manufacturer		Schneider	Confirmed	3	0
Model		V321-BBAAD-AAANA-A1	Confirmed	3	0
Power Supply		24VDC		1	0
Communications		Via Modbus TCP Communication module		1	0
Design note		Supplier to design, product selection, install and commission Arc fault Detection system come complete with minimum of 2x fast acting trip contacts rated for a minimum of 24VDC operating voltage. Current Supervision required for all incomers.	Confirmed	3, 7	0
Sensor Locations		Incomer and all outgoing modules including cable ways. Arc fault monitoring is not required for areas (E.g. busbar) where the component design is classified as an arc free zone in accordance to IEC61641.		1	0
Current Transformer		As per manufacturer requirements. To be installed by Switchroom Vendor in the Transformer room.		1	0
VSD Feeders					
Feeder Type		Moulded Case Circuit Breaker (MCCB)		1	0
Fixed / Demountable / Withdrawable		Demountable		1	0
MCCB Make		Terasaki	Confirmed	3	0
MCCB Model (series)		Tembreak Pro	Confirmed	3	0
MCCB Protection		Adjustable Magnetic and Thermal as standard, rating as per manufacturer's protection table		1	0
Outgoing cable connection (cable(s) exiting MCC)		VTA - Onto MCCB or terminals as per manufacturer's recommendation	Confirmed	3	0

Description	Unit	Duty	Vendor	Notes	Rev
Neutral Connection		Provided inside feeder module via Neutral busbar dropper		1	0
VSD Starter Control Voltage		24V DC		1	0
DOL Motor Starters					
Feeder Type		Moulded Case Circuit Breaker (MCCB)		1	0
Fixed / Demountable / Withdrawable		Demountable		1	0
MCCB Make		Terasaki	Confirmed	3	0
MCCB Model (series)		Tembreak Pro	Confirmed	3	0
MCCB Protection		Adjustable thermal magnetic as standard, rating as per manufacturer's Type 2 coordination chart for IE3 high efficiency motors. Adjustable earth leakage and electronic trip units as shown on schematics.		1	0
Contactor Model		Type 2 Coordinated AC3 duty suitable for IE3 high efficiency motors	Confirmed	3	0
Electronic Overload Relay (EOL)					
Electronic Overload Make / Model		Rockwell E300 series		1	0
Electronic Overload functions		As a minimum: - Current sensing Class 10 motor thermal protection I/O modules (If required) - PTC Thermistor input Model included can exclude Earth Leakage or Ground Fault protection as this is provided by the Earth Leakage relay	Confirmed	3	0
Electronic Overload Communication		Modbus TCP – Anybus-E300-MBTCP Comms Module		1	0
Electronic Overload Expansion I/O module		Where required		1	0
Sensing module		Integral or External to suit motor duty		1	0
HMI/ Starter Control station module		Required		1, 7	0
Thermal Overload Model (applicable to multi-motor DOL Starters)		VTA - Selected to suit Type 2 coordination tables for AC3 duty for high efficiency motors	Confirmed	3	0
Outgoing cable connection		Onto contactor, terminals, or TOL as per manufacturer's design drawings		1	0
Neutral Connection		Provided inside feeder module via Neutral busbar dropper		1	0
Motor Starter Contactor Coil Voltage		240V AC		1	0
Motor Starter Control Voltage		24V DC		1	0
MCCB Feeders					
Feeder Type		Moulded Case Circuit Breaker (MCCB) with External Earth Leakage		1	**0**
Fixed / Demountable / Withdrawable		Demountable		1	0
Feeder Make		Terasaki	Confirmed	3	0
Feeder MCCB Model		Tembreak Pro	Confirmed	3	0
MCCB Protection		Adjustable thermal magnetic as standard, 3 Pole with Solid Neutral sensing, adjustable earth leakage and electronic trip units where shown on SLD. VTA range of adjustment offered for thermal/ long time trip.	Confirmed. Earth leakage provided by external relay and toroid	3	0
Fault limiting fuses		Yes, where specified on standard drawing.		1	0
Outgoing cable connection		Onto terminals as per manufacturer's design drawings		1	0
Neutral Connection		Provided inside feeder module via Neutral busbar dropper		1	0
Standard Motor Starter and VSD Feeder Sizes					
DOL motor starters, VSD feeders and soft starter motor starters grouped into standard sizes for each starter type with all starters of the same type and size using the same components	Yes / No	Yes		1,5	0
Nominated Standard Sizes		Vendor to advise standard module sizes, Engineer standard schematics will be adjusted to suit	TBA	3,5	0
Control Power Supply & Distribution					
24V DC Distribution		Redundant 240VAC/24VDC power supplies fed from switchroom 240VAC UPS DB		1	0
Make / model		PHOENIX QUINT4-PS/1AC/24VDC/20		1	0
Separate 24V DC Circuit Breakers for each tier	Yes / No	Yes, located at top of cable zone		1	0

Description	Unit	Duty	Vendor	Notes	Rev
Inter-Panel Wiring					
Routing		Routed in cable duct running along top of MCC tiers	Confirmed	3	0
Segregation		Ethernet Comms to be segregated by metallic barrier of in separate duct	Confirmed	3	0
Duct Space for Future Expansion		Minimum 20% space to be provided in control wiring ducts for connection of future MCC modules.		1	0
Communication					
MCC Network Switch (free issued)		Installed in the top of tier cable zone. Maximum of 1 per two tiers	N/A	3	0
Labelling					
Installation Details	Refer standard electrical drawings 2000-F00-STD-EL-10009 and 2000-F00-STD-EL-10010			1	0
Switchboard Label	Etched traffolyte label, located on incomer tier, with MCC number, MCC description and SLD number at minimum.			1	0
Switchboard Nameplate Label	Etched traffolyte nameplate label located on incomer tier. Label shall include the switchboard ratings, serial number, and date of manufacture.			1	0
Tier Label	Etched traffolyte label (black on white background) to be located on the top of each tier, with the MCC tier number.			1	0
MCC Module Outer Door Label	Etched traffolyte label (black on white background) on the outer door of each MCC module. Each label shall contain the module tier position, drive kW rating or feeder Amp rating, equipment tag number, equipment description and the MCC schematic drawing number.			1	0
MCC Module Inner Labelling	Etched traffolyte label (black on white background) with equipment number for equipment internal to MCC module (e.g. MCCB, MCB, Relays, Terminal Strips).			1	0
Terminals and Panel Wiring	Terminals to be numbered per design drawings, internal wiring shall be marked at both ends as per design drawings.			1	0
Miscellaneous Components					
Miniature Circuit Breakers Make & Model		NHP DIN, 10kA Type	Confirmed	3	0
MCCB Handles	Yes / No	Yes, interlocked with cubicle doors. Padlockable both inside and outside of module door		1	0
Control Relays		NHP Finder Type	Confirmed	3	0
Control and Power Terminal Make & Model		Weidmuller, WDU series	Confirmed	3	0
Earth Terminal Make & Model		Weidmuller, WPE series	Confirmed	3	0
LED Indicator Lamps & Pushbuttons		IPD P9 Series	NHP	3	0
Voltage Test Terminals (MCC Incomer Cubicle)		Weidmuller, WTL series	Confirmed	3	0
Earth Leakage Relay – VSD Feeder	Make/ Model	Ampcontrol/ELV		1,7	0
Earth Leakage Relay – All other motor starter/feeder	Make/ Model	NHP/RD1EP21H		1,7	0
Test Requirements					
Type (Verification)		Required Provide test certificates		1	0
Routine		Required Including: - Insulation Test of MCC Bus - Functional checks of all MCC modules - ACB incomer function testing - Primary & Secondary injection testing of AHF and MEN CTs		1	0
Shipping and Delivery					
Packaging		Installed in switchroom		1	0
Dimensions	mm(L) x mm(W) x mm(H)		4920 x 1000 x 2329	2	0
Weight	kg		4000	2	0
Shipping Section Dimensions	mm(L) x mm(W) x mm(H)		2800 x 1000 x 2329	2	0
Heaviest Shipping Section Weight	kg			2	0

Notes
1. Fixed. The Vendor shall comply with this data, as a minimum. Vendor shall state their applicable ratings/data or compliance.
2. Vendor to Advise.
3. Preferred, Vendor to advise and provide reasoning for difference.
4. Note deleted.
5. Vendor to nominate standard module sizes.
6. Vendor to provide at tender stage an initial AS61439.1 Appendix D.1 Statement of Compliance (SoC) document for the complete assembly with final SoC provided after testing. Refer to Attachment A.

Notes
7. Where provided, EOL HMI, Arc flash protection relay, power meter, Earth Fault Relays or Earth Leakage Relays shall be door mounted to access push buttons, dip-switch settings and to view indications without sacrificing MCC fault rating and arc flash containment. One arc flash sensor should be placed in each feeder module and bus zone tier.

Vendor Drawing and Document Requirements List (VDDR)

E – Electronic Copy H – Hard Copy MDR – Manufacture's Data Report

FAT – Factory Acceptance Testing IOM – Installation and Operating Manual

VDDR			Required with Tender	Required after Award		As Built Required
			Copies	Copies	Weeks	Copies
A		**Production Schedule**				
	A01	Production Schedule / Bar Chart		1E	2	
	A02	Monthly Report		Monthly		
B		**Registers and General Drawing Requirements**				
	B01	Drawing Schedule / Register (Drawings Expected)		1E	2	
	B02	General Arrangement with Foundation Loads (in AutoCAD 2010 and PDF)		1E	6	1E
	B03	Cross-section		1E	6	1E
C		**Mounting and Erection Drawings**				
	C01	Foundation and Mounting Details (Provided in B02)				
	C02	Pipe Spools				
	C03	Outline Dimensions, Cut-out and Support Details (Provided in B02)				
	C04	Match Marked Erection Drawings				
	C05	Motor GA (with Fixing and Shaft Details, and Overall Dimensions)				
D		**Electrical Drawings (in electronic form as appropriate)**				
	D01	Single Line Diagrams				
	D02	Schematic Drawings				
	D03	Terminations / Loop Diagrams				
	D04	Panel Layout and Fabrication Details		1E	6	1E
	D05	Nameplate		1E	6	1E
	D06	Cable Block Drawings				
	D07	Switchboard GAs (Overall, Incomer and Module Type)		1E	6	1E
E		**Process and Piping Drawings**				
	E01	Process Flow Sheets				
	E02	Process and Instrument Diagram				
	E03	Pneumatic or Hydraulic Diagram				
	E04	Lubrication Diagram				
F		**Technical Data**				
	F01	Equipment Information Schedule	1E			
	F02	Electrical Performance data sheet		1E	6	
	F03	Equipment List		1E	6	1E
	F04	Vendor's Surface Protection Details		1E	6	
	F05	Lubrication Data Schedule				
	F06	Material Schedule				
	F07	Bolt Schedule				
	F08	Fan Performance Curves				
	F09	Pump Performance Curves (with Duty Points Marked)				
	F10	Cable Schedule				
	F11	Vendor Catalogue Information	1E			
	F12	Calculations				
	F13	Completed $_{\text{Engineer}}$ Datasheets	1E			
	F14	Motor Datasheets				
	F15	Electrical Equipment List				
	F16	Instrument List				
	F17	I/O List				
	F18	Communications Addressing and Signal Mapping List				

VDDR			Required with Tender	Required after Award		As Built Required
			Copies	Copies	Weeks	Copies
G		**Inspection, Certification and MDR**				
	G01	Inspection and Test Plans		1E	6	
	G02	Inspection and Test Procedures (generic)				
	G03	Quality Plans and Procedures (generic)		1E	4	
	G04	Manufacturer's Data Report (MDR) – Index / Contents		1E	4	
	G05	Manufacturer's Data Report (MDR)		1E	Delivery -4	
	G06	Material Certificates		In MDR		
	G07	Statutory Authority Certificates		In MDR		
	G08	Test Data and Other Certificates		In MDR		
	G09	Welding Procedure / Specification				
	G10	Non-Destructive Testing (NDT) Procedures				
	G11	Material Safety Data Sheets				
	G12	Commissioning Plan and Procedure		1E	Delivery - 4	
	G13	Structural Design Certification				
	G14	Performance Guarantee Statement				
H		**Shipping and Packing Details**				
	H01	Material / Shipping List	1E	1E	Delivery -4	
	H02	Package Type and Quantity (Carton, Case, Crate)		1E	Delivery -4	
	H03	Shipping Packages – Weights and Dimensions		1E	Delivery -4	
	H04	Shipping and Handling Procedure		1E	12	
J		**Manuals**				
	J01	Installation, Operation and Maintenance Manual (Preliminary)		1E	Delivery -4	
	J02	Installation, Operation and Maintenance Manual (Final)		1E	Delivery	
K		**Spare Parts Information**				
	K01	Spare Parts Catalogue and Recommended Spares List (Copy also in Manuals)	1E	1E	8	
	K02	Recommended Spare Parts List with Pricing (Not Included in Manuals)	1E	1E	Prelim 4 Delivery +1	
	K03	Special Tools list	1E	1E	8	
M		**General**				
	M01	Control Philosophy and Logic Diagram				
	M02	Functional Description				

ATTACHMENT A: VERIFICATION / STATEMENT OF COMPLIANCE WITH AS/NZS 61439.1

Supplier to complete the statement of compliance table below and provide relevant verification documentation with response.

No.	Characteristic to be verified	Clauses[1]	Statement of Compliance	Verification type[2]	Documentation attached
1	Strength of material and parts:	10.2			
	Resistance to corrosion	10.2.2		Test	103854 / 104031
	Properties of insulating materials:	10.2.3		Assessment	
	Thermal stability	10.2.3.1		Test	103854 / 104031
	Resistance to abnormal heat and fire due to internal electric effects	10.2.3.2		Test	103854 / 104031
	Resistance to ultra-violet (UV) radiation	10.2.4		Outdoor Switchboards only	103854 / 104031
	Lifting	10.2.5		Test	103854 / 104031
	Mechanical impact	10.2.6		N/A	
	Marking	10.2.7		Test	103854 / 104031
2	Degree of protection of enclosures	10.3		Test	103854 / 104031
3	Clearances	10.4		Test	103854 / 104031
4	Creepage distances	10.4		Test	103854 / 104031
5	Protection against electric shock and integrity of protective circuits:	10.5			
	Effective continuity between the exposed conductive parts of the assembly and the protective circuit	10.5.2		Test	103854 / 104031
	Short circuit withstand strength of the protective circuit	10.5.3		Test	103854 / 104031
6	Incorporation of switching devices and components	10.6		Assessment	
7	Internal electrical circuits and connections	10.7		Assessment	
8	Terminals for external conductors	10.8		Assessment	
9	Dielectric properties;	10.9			
	Power-frequency withstand voltage	10.9.2		Test	103854 / 104031
	Impulse withstand voltage	10.9.3		Test	103854 / 104031
10	Temperature rise limits	10.10		Test	103854 / 104031
11	Short circuit withstand strength	10.11		Test	103854 / 104031
12	Electromagnetic compatibility (EMC)	10.12		Test	103854 / 104031
13	Mechanical operation	10.13		Test	103854 / 104031

[1] Refers to relevant clause or subclause in AS/NZS 61439.1.
[2] Refer to Table D.1 of AS/NZS 61439.1 for verification options.